

How river structure disperses and broadens the same DOC pulse

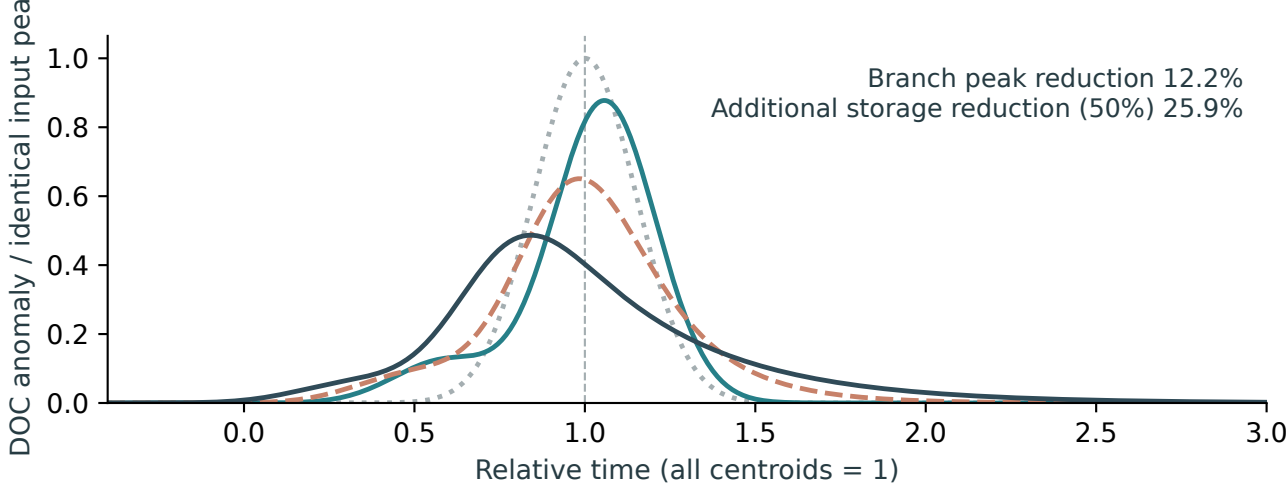
..... Equal branches, translation - - - Actual branches, 50% common-time storage
— Actual branches, translation — Actual branches, 100% common-time storage

a Lower dispersion / lower common share

Mapped common-trunk waterbody path 0.0%

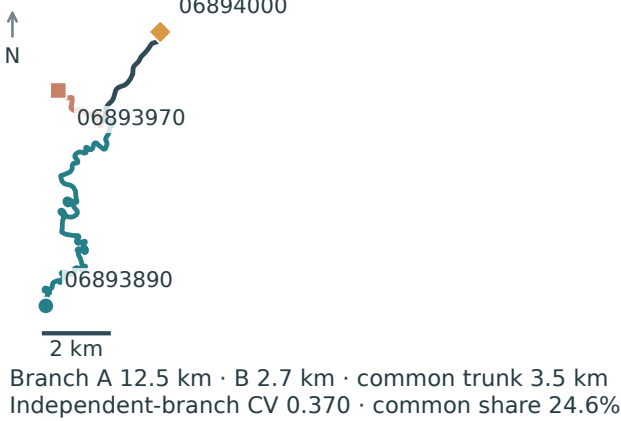


Same mean arrival, different response shapes

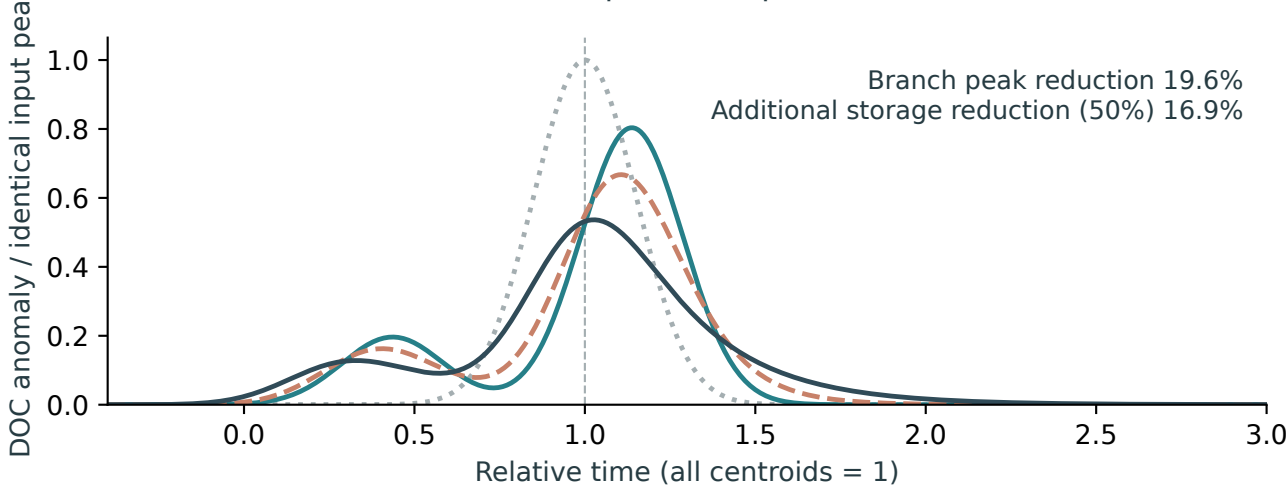


b Higher dispersion / lower common share

Mapped common-trunk waterbody path 0.0%

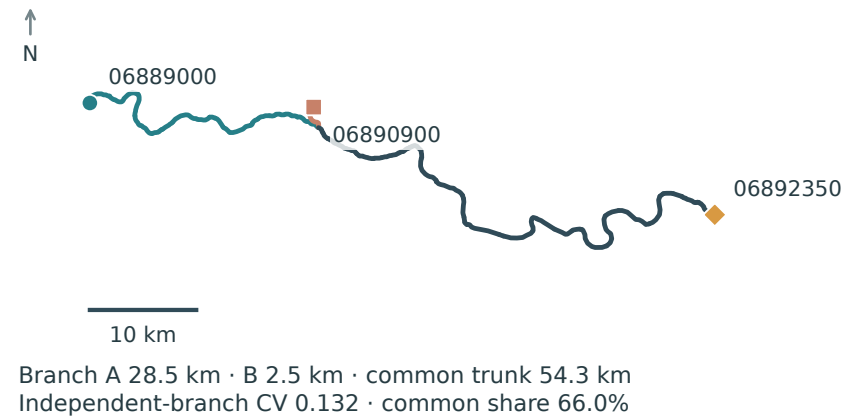


Same mean arrival, different response shapes

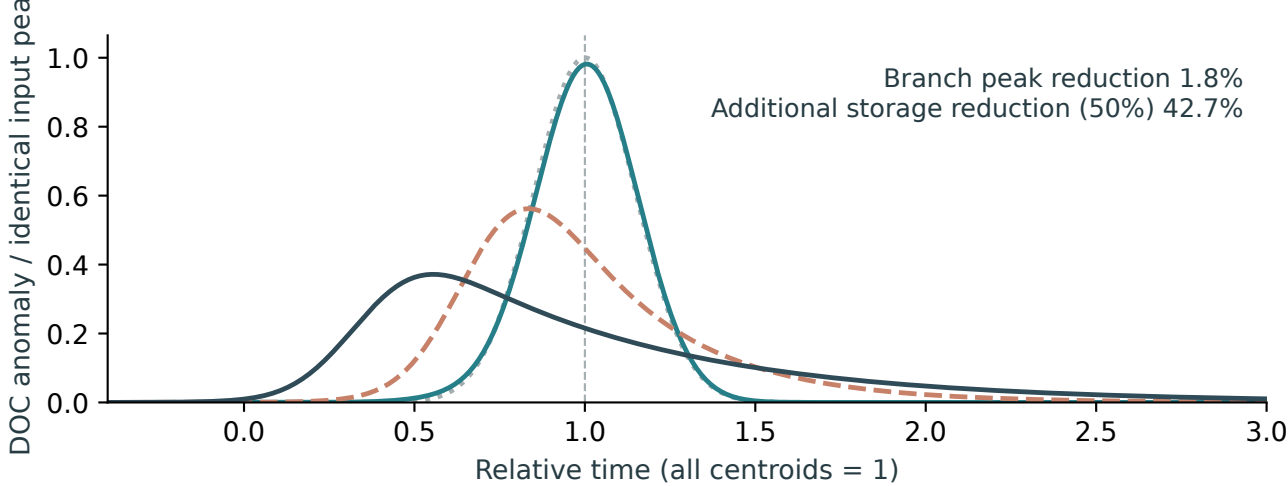


c Lower dispersion / higher common share

Mapped common-trunk waterbody path 0.0%

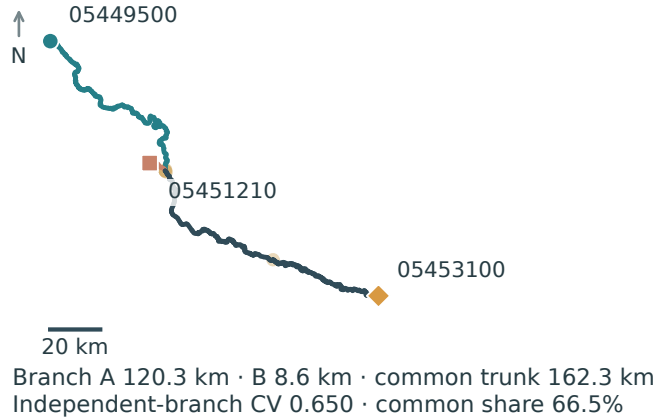


Same mean arrival, different response shapes

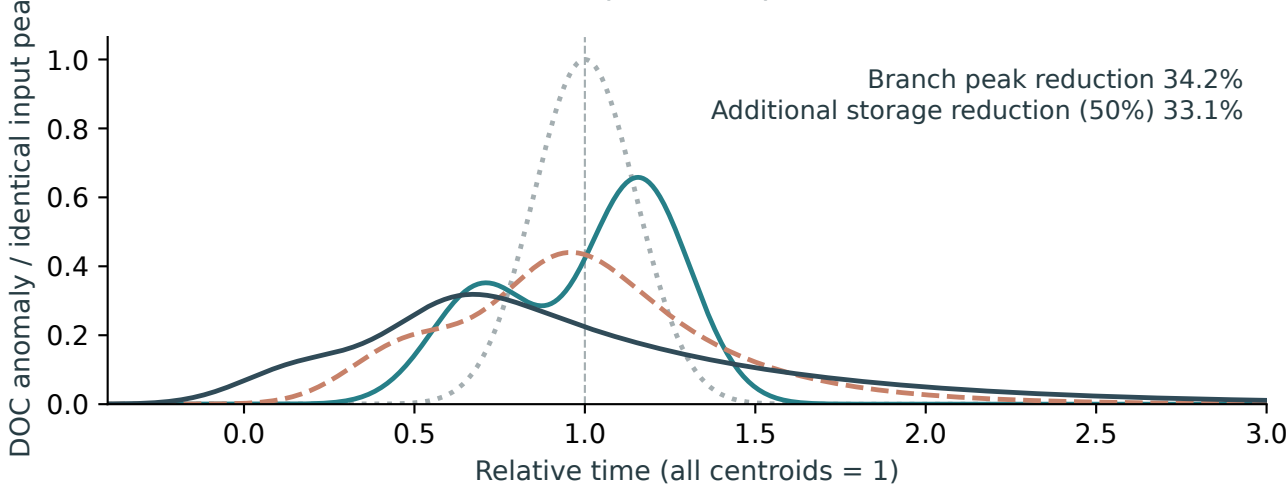


d Higher dispersion / higher common share

Mapped common-trunk waterbody path 0.5%

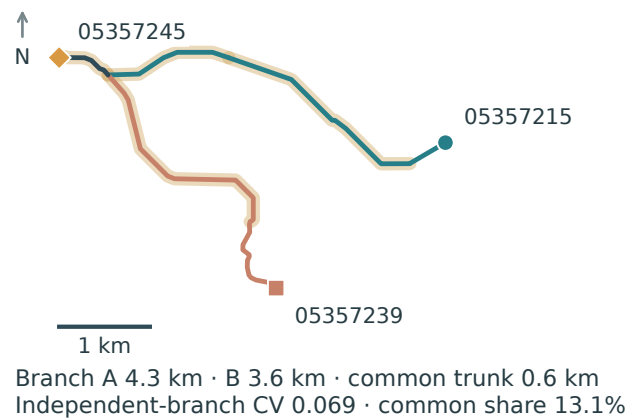


Same mean arrival, different response shapes

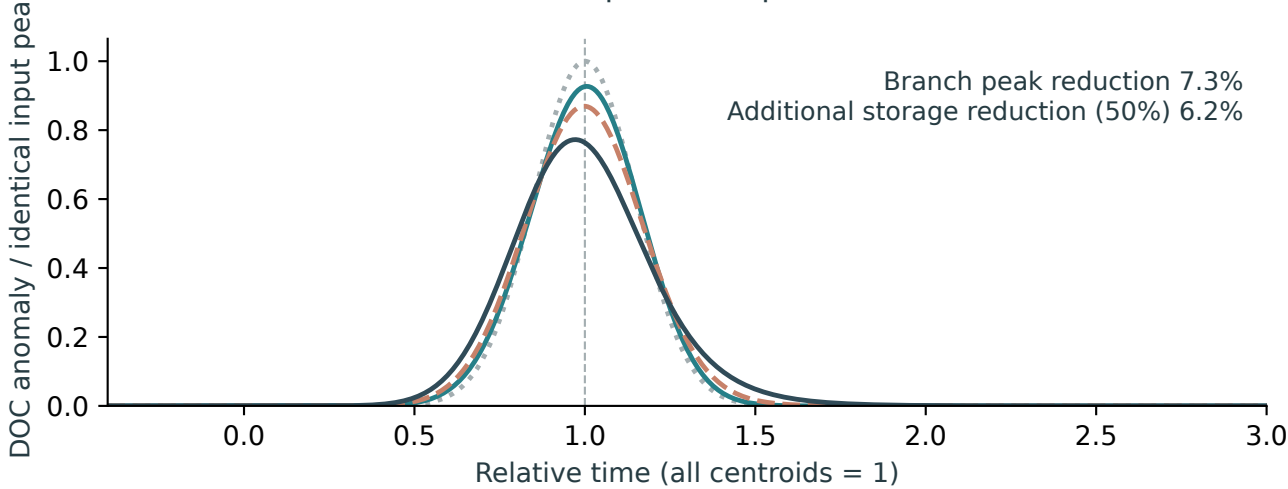


e Highest common-trunk waterbody share

Mapped common-trunk waterbody path 89.2%



Same mean arrival, different response shapes



Left: real NHDPlus channels; teal branch A, orange B, navy common trunk; gold underlay marks waterbody reaches. Independent scales.
Right: identical forcing, shares and mean arrival; input SD = 0.15. Storage allocates common time, not mapped waterbody length.
Relative time is not days; these are simulated conservative responses, not measured DOC reductions.